

An intelligent fuzzy-based system for handover decision in 5G-IoT networks considering network slicing and SDN technologies

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ABSTRACT

By enabling 5G technology, IoT networks can improve the performance of connected IoT devices. However, handover, ping-pong effect and load balancing are critical issues because of massive and unplanned deployment of small cells. The handover operation in 5G wireless networks is complicated because there are many radio access techniques and technologies. Therefore, the mobility management is required to provide QoS for various applications. In this paper, we present a fuzzy-based system for handover decision in 5G wireless networks considering different network slicing parameters. We implement two Fuzzy-based Handover Decision Models (FHDM): FHDM1 and FHDM2. We evaluate the proposed models by simulations. For FHDM1, Slice Delay (SD), Slice Bandwidth (SB), and Slice Load (SL) are three input parameters and Handover Decision (HD) is the output parameter. For FHDM2, Slice Reliability (SR) is a new considered parameter, so it has four input parameters and the output parameter is HD the same as FHDM1. We found from simulation results that the considered parameters have different effects on HD. For both models, the HD value increases as SD and SL values increase while HD value is decreasing when SB and SR values increase. The results of the simulation indicate that although FHDM2 is more complex than FHDM1, it performs better HD since considers four input parameters. From the simulation results of FHDM2, when we changed SD value from 10% to 90%, we found that the HD value is increased by 50% when SR is 50% and SL is 90%. This indicates that when a user is connecting with the present slice that has bad QoS, it should handover to another slice with better QoS.

1. Introduction

The 5th Generation (5G) mobile communication is becoming challenging and complicated by the rapid growth of mobile devices. Therefore, the development of network technologies and network devices are expected to satisfy users Quality of Service (QoS) [1–4]. Mobility management is the critical operation for improving QoS and providing uninterrupted services to users on 5G wireless networks. Also, users require a seamless handover with high efficiency, low latency, and high data rates when they move with different mobility [5].

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Network Slicing (NS) is a 5G wireless network technology used to meet consumers' QoS demands. The NS is a unique technology that creates new application services over the same physical infrastructure utilizing Software-Defined Networks (SDNs) and Network Function Virtualization (NFV). It can support dependable network services on demand with less resource usage. Therefore, the NS is very important for handover in 5G wireless networks. Inter-slice handover is a new type of handover that consider multiple slice services rather than just the User Equipment (UE) physical mobility when performing handover [6].

Because traditional networks have complicated reconfiguration and they are difficult to manage, SDN is one of the new approaches that can improve network performance to face new difficulties and challenging problems in 5G wireless networks. There are many issues and user demands, therefore technologies suitable for 5G wireless networks need to be developed.

Designing and engineering 5G wireless networks is not an easy task because require the optimization of many parameters. In [7], the authors design a framework considering NS for ultra-low latency scenarios. However, these parameters are often uncorrelated and therefore a trade-off on the solutions is required in the optimization process. These parameters include connectivity, coverage, Quality of Service (QoS), the network cost and so on. Also, there are different approaches for traffic control and engineering in 5G wireless networks such as Call Admission Control (CAC), Selection of Radio Access Technologies (RATc) and handover. During these processes, there are more than two uncorrelated parameters for optimization, thus the problem becomes an NP-hard problem, which is a problem that cannot be solved in polynomial time. Recently, there are different approaches such as machine learning, AI, soft computing and evolutionary-based algorithms which are capable of solving NP-hard problems.

In our previous work [8–10], we presented some Fuzzy-based systems for CAC in 5G wireless networks. In order to make the handover decision in 5G wireless networks are required many uncorrelated parameters, which makes the problem NP-Hard. Also, the decision should be done in real-time. For this reason, intelligent and heuristic approaches should be considered. Therefore, in this paper, we propose a Fuzzy-based Handover Decision System (FHDS). We present and compare two Fuzzy-based Handover Decision Models (FHDMs): FHDM1 and FHDM2. The FHDM1 considers three parameters and FHDM2 considers four parameters by adding Slice Reliability (SR) as a new parameter.

The contributions of this work are as follows.

- We investigated and proposed different parameters for handover decision in 5G wireless networks.
- We propose and implement an intelligent system for handover decision in 5G wireless networks based on FL.
- We implement two models to evaluate the proposed system.
- We compare the simulation results for both models for different scenarios.
- FHDS2 can provide better handover decision, because it considers 4 input parameters.

The rest of the paper is organized as follows. Section 2 present Related Work. In Section 3, we introduce 5G-IoT networks. A description of SDN is provided in Section 4. Network Slicing is presented in Section 5. We introduce a short overview of FL in Section 6. In Section 7, we present the proposed fuzzy-based system and its models. In Section 8, we discuss the simulation results of two models. Section 9 gives conclusions and future work.

2. Related work

In this section, we present some of handover related work for SDN and 5G wireless networks. We also present some related work considering network slicing and inter-slice handover methods. In order to have an effective inter-slice handover procedure, it is important to choose an appropriate target slice promptly and trigger handover to the target slice.

Many research works consider technologies such as SDN, NFV, Fuzzy Logic (FL) and intelligent algorithms. For example, in order to reduce the processing delay in the handover procedure, the QoS can be enhanced by applying FL in SDN [11–13]. However, in these papers, the authors consider only conventional input parameters such as Received Signal Strength (RSS), data transmission rate, battery charge or packet loss rate and delay.

In [14], the slices service resources is maintained during handover procedure using three heuristics algorithms: Simple Algorithm, Greedy Handover Algorithm and Intelligent Handover Algorithm. The handover procedure can maintain the required services for each slice and the availability of slices when the handover will be activated. The handover problem is investigated also in [15] using reinforcement learning techniques. The authors make the assumption that a separate subset of BSs supports each slice. The goal is to minimize the long-term handover costs while maintaining user QoS. The handover cost is defined from the cost of switching NS, the cost of switching BS, the cost of switching BS and NS, the cost of creating a new NS.

For e-Health 5G Use Cases, in [16], the authors introduced the Cloud-Network-Slicing Mobility Control (CNS-MC) approach to maintain the m-health UE experience during mobility procedure.

For V2X environment, the authors implement a V2X Slice Selection Function (SSF-V2X) as a SDN application to trigger inter-slice handover by considering service requirements and network constraints [17]. Also, in [18], the authors proposed a inter/intra slice handover management architecture called Connection Mode as A Service (CMaaS). The CMaaS controller is used for the Inter-Slice Handover and Skipping Mode as A Service (SMaaS) is used for the Intra-Slice Handover to minimize the handover cost.

In [19], the authors consider inter-slicing handover security and reliability. They address security flaws in 5G network slicing from attack detection and localization algorithms that launch a Distributed Denial of Service (DDoS) during inter-slicing procedure to find the weaknesses point. During the inter-slice handover phase, the lengthy authentication process is abused to start a DDoS assault.

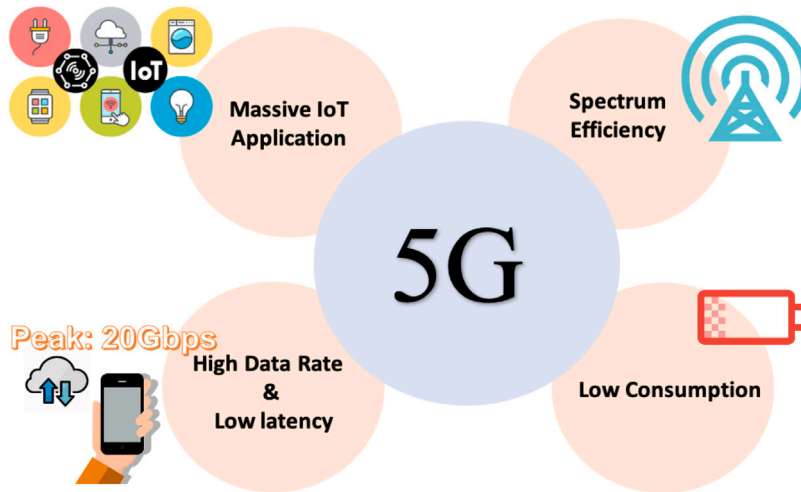


Fig. 1. 5G IoT challenging.

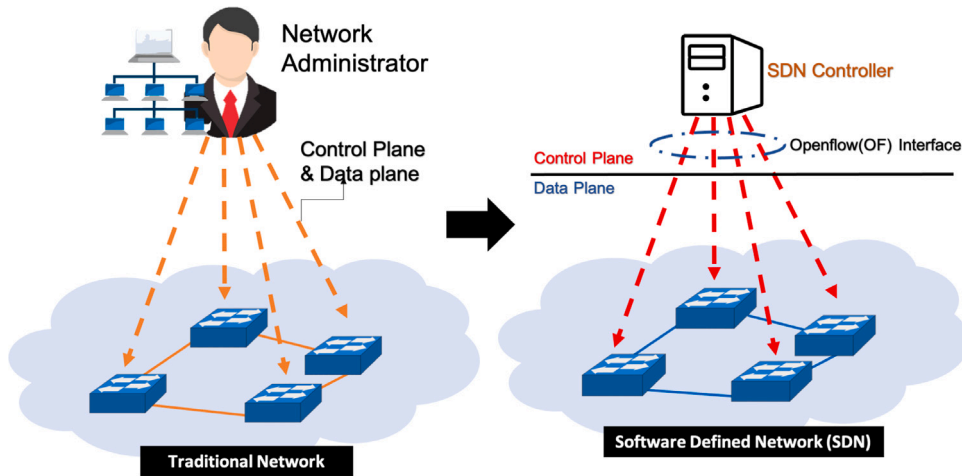


Fig. 2. Comparison of traditional network and SDN.

3. 5G-IoT networks

The number of connected devices is rapidly increasing, including Internet of Things (IoT) devices that cause the increasing demands of applications [20–22]. By enabling 5G technology, such as SDN, NS, Machine-to-Machine (M2M) communications, massive MIMO and millimeter-wave (mmWave) technologies, IoT networks can improve the performance and reliability of connected IoT devices. Also, 5G-IoT deployment will generate a diverse form of traffic, reliability, bit rates, energy consumption, security and privacy for improving Quality of Service (QoS) requirements and achieving massive Machine-Type Communication (mMTC) as shown Fig. 1 [23].

Recently, many research works deal with design of systems appropriate for 5G-IoT networks. One example is the utilization of NS for managing a large number of heterogeneous IoT network slices dynamically [24]. Also, 5G-IoT infrastructure implement SDN/NFV for improving load balancing, fault tolerance and congestion avoidance in IoT network [25]. In addition, massive communication interfaces issues (redundant communication capability problem and waste energy) can be improved by 5G massive MIMO [26].

4. Software-defined networks (SDNs)

The SDN can optimize and simplify network operation. SDN implements a SDN controller or a logically centralized network control into the network architecture. The traditional network and SDN techniques are compared in Fig. 2. The traditional network relies on only physical infrastructure, making management and control of traffic network and network devices difficult and complex. Furthermore, because each device requires its own control and operation, the network configuration is hard to configure and

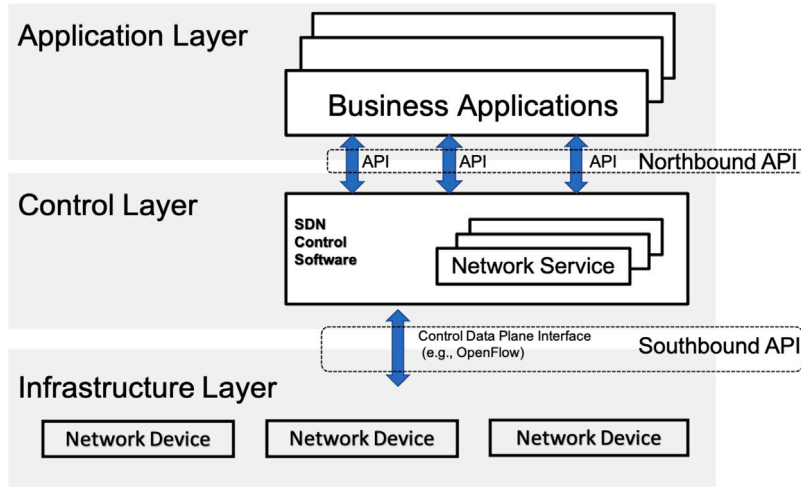


Fig. 3. Overview of SDN architecture.

reconfigure all network devices quickly. On the other hand, SDN builds a virtualized control plane with smart management, when network managers utilize a virtualized controller to control and alter the entire network [27–32]. The SDN structure is shown in Fig. 3 and explained in following.

- **Infrastructure Layer** consists of network equipments, such as switches and routers. It follows the SDN controller's commands and transfers data into network devices.
- **Southbound Interfaces** are the connecting bridges between the Infrastructure Layer and the Control Layer. OpenFlow is the most widely used and recognized protocol in this interface. The protocols enable the Control Layer to design new forwarding plane policies.
- **Control Layer** consists of controllers that receive information or policies from the Application Layer and send analyzed information or order to the lower layer.
- **Northbound Interfaces** offer various Application Programming Interfaces (APIs).
- **Application Layer** gathers network information from the controller for decision-making and creates a conceptual representation of the network.

By enabling new services, the SDN can control and improve the network performance. The SDN can adapt to traffic congestion situations, enabling users to quickly manage and allocate resources across the control plane. Mobility management between radio access technologies such as 5G, 4G, and Wifi becomes simpler and faster [33]. The 5G handover procedure can be straightforward and the delay can be minimized by implementing SDN [34]. Also, SDN has been applied in different networking scenarios, including commercial and industrial networks. Nevertheless, the complete network architecture must be modified in order to use the SDN controller and protocol.

5. Network slicing

The NS separates a single hardware service into several logical network services known as “Slices”. Thus, the virtual logical networks are undisturbed from other slices because each of them is logically independent. As a result, NS provides enhanced QoS and flexibility to users service demands compared to traditional networks. In addition, because of the non-interfering of other slices, the security and dependability of slices can be enhanced [35–38].

The Next Generation Mobile Networks (NGMN) is developing 5G NS concept as shown in Fig. 4. There are three primary layers to the NS procedure [39–41].

- **Service Instance Layer** provides end to end services or commercial services offered by an application provider or mobile network operator.
- **Network Slice Instance Layer** comprises various network resources and functions that give Service Instances, which are optimized network characteristics by following user requirements.
- **Resource Layer** consists of both physical and logical components that are managed by the Network Slice Instance layer.

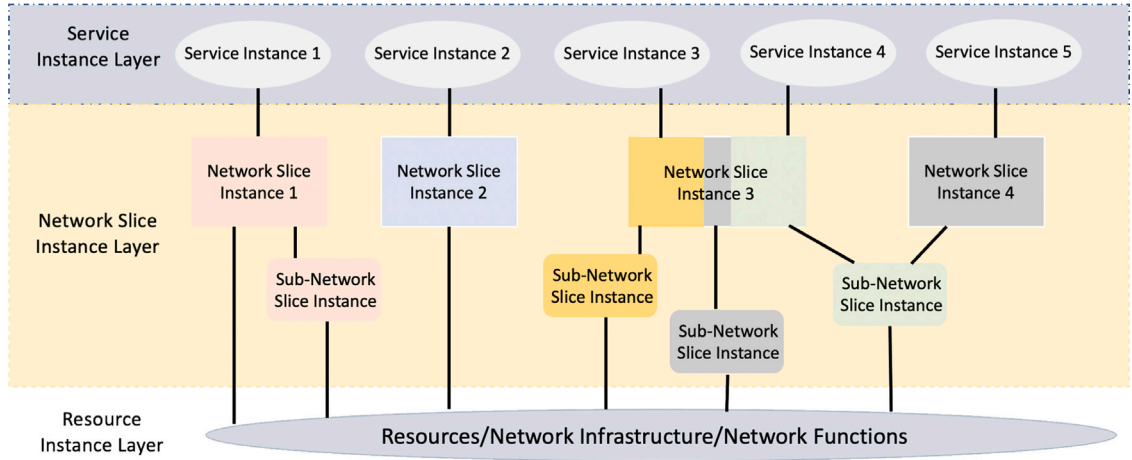


Fig. 4. NGMN NS concept.

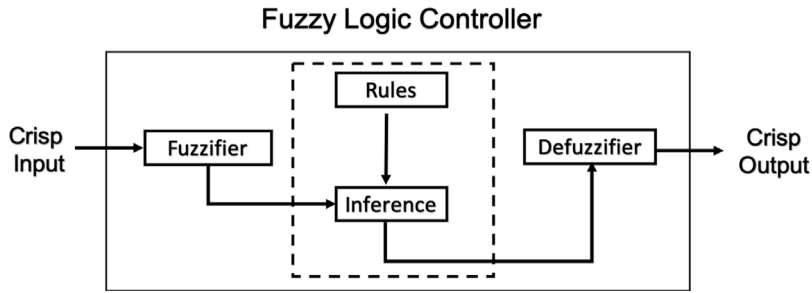


Fig. 5. Structure of Fuzzy Logic Controller (FLC).

6. Overview of Fuzzy logic (FL)

A FL system considers complete interval statements between False (0) and True (1) to describe human reasoning by conversion of vector data (input) into scalar data (output). The FL is employed in several systems, including TV picture correction, aviation control (Rockwell Corp.) and Sendai subway operating (Sony) [42–44].

Fig. 5 shows the structure of FLC, which has four components.

- **Fuzzifier** converts the crisp input into identical membership values of the relevant fuzzy sets by using the membership functions. The membership function specifies how to generate a graph by mapping each point in the crisp inputs with a degree of membership ranging from 0 to 1. In general, a single input variable can be described by multiple membership functions.
- **Fuzzy Rules** could be given by a professional or can be taken from numerical data. A set of IF-THEN expressions can describe the engineering rules. For example, if x and y are linguistic variables defined by the fuzzy sets A and B , the expression will be “IF x is A , THEN y is B ”. The control (input) variable is called x , while the solution (output) variable is called y . Conditional sentences use the form “IF... THEN”. It consists of “IF” (also known as the antecedent) and “THEN” (also known as the resultant).
- **Inference Engine** uses these rules to compute a fuzzy output (the results). There are many fuzzy logic inferences such as Mamdani fuzzy inference engine, Takagi–Sugeno fuzzy inference engine, and so on.
- **Defuzzifier** converts a fuzzy output (the result) into a crisp output (the outcome), which is used for control.

7. Proposed Fuzzy-based system

In this section, we present our proposed fuzzy-based system. In Fig. 6, we show the overview of our proposed system. Each evolve base station (eBS) will get orders from the SDN controller, allowing these stations to communicate and relay data to UE. Also, each eBS contains a number of slices with various services. By gathering the data based on network traffic status, the proposed Fuzzy-based system implemented in the SDN controller will manage inter-slice handover. The SDN controller will offer a communication channel between eBS and the 5G core network.

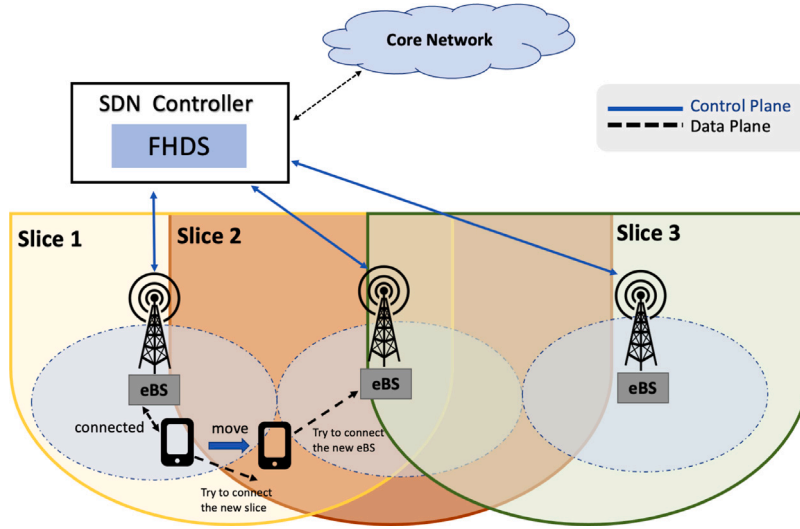


Fig. 6. Proposed system overview.

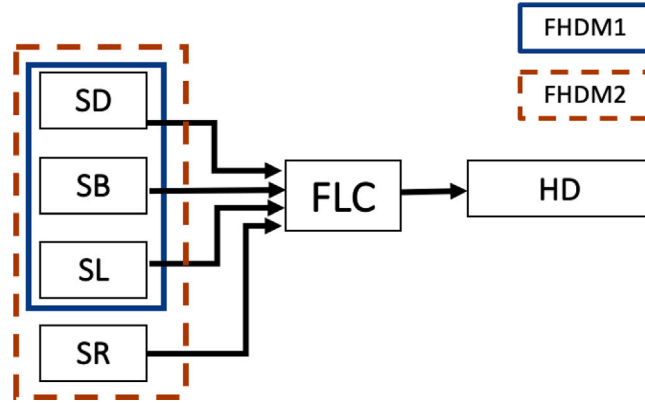


Fig. 7. Proposed system structure.

The proposed system is called Fuzzy-based Handover Decision System (FHDS) in 5G Wireless Networks. The structure of FHDS is shown in Fig. 7. We implement two Fuzzy-based Handover Decision Models (FHDMs): FHDM1 and FHDM2. The FHDM1 considers three input parameters: Slice Delay (SD), Slice Bandwidth (SB), Slice Load (SL) and the output parameter is Handover Decision (HD). In FHDM2, we consider Slice Reliability (SR) as a new parameter. Thus, FHDM2 has four input parameters.

We explain four input parameters and the output parameter in following.

Slice Delay (SD): High queueing and network delay are caused by the slice delay. Also, the impact of delay may lead to rejection of demands, resulting in low slice resource utilization. As a consequence, the handover procedure is required to satisfy the QoS.

Slice Bandwidth (SB): The available bandwidth for a slice is known as slice bandwidth. The bandwidth resources can affect the number of bytes a slice can transmit at a given time. The SB should be assigned according to the channel variety and various bandwidth needs of the service types in different slices. The possibility of a handover will decrease as the SB is increased.

Slice Load (SL): A slice becomes heavily loaded when its resources are heavily utilized by a large number of users. So, the user will handover to another slice with a lower load to have better service.

Slice Reliability (SR): The SR is derived from the error rate caused by network traffic congestion or some network node failures. The user will make handover to alternative slices when a slice has low reliability.

Handover Decision (HD): The HD parameter is used for making the decision for the handover process. By many simulations, we found that 7 Levels are good enough for our system, where HD1 indicates the extremely low possibility of inter-slicing handover and HD7 indicates the extremely high possibility of inter-slicing handover.

The membership functions are shown in Fig. 8. The proposed fuzzy system can be used in a wide variety of applications. We use Delay, Bandwidth, Load and Reliability numbers between 0 and 100% to simply fuzzify them. We consider triangular and trapezoidal membership functions because they are more appropriate for real-time operations. We show parameters and their term

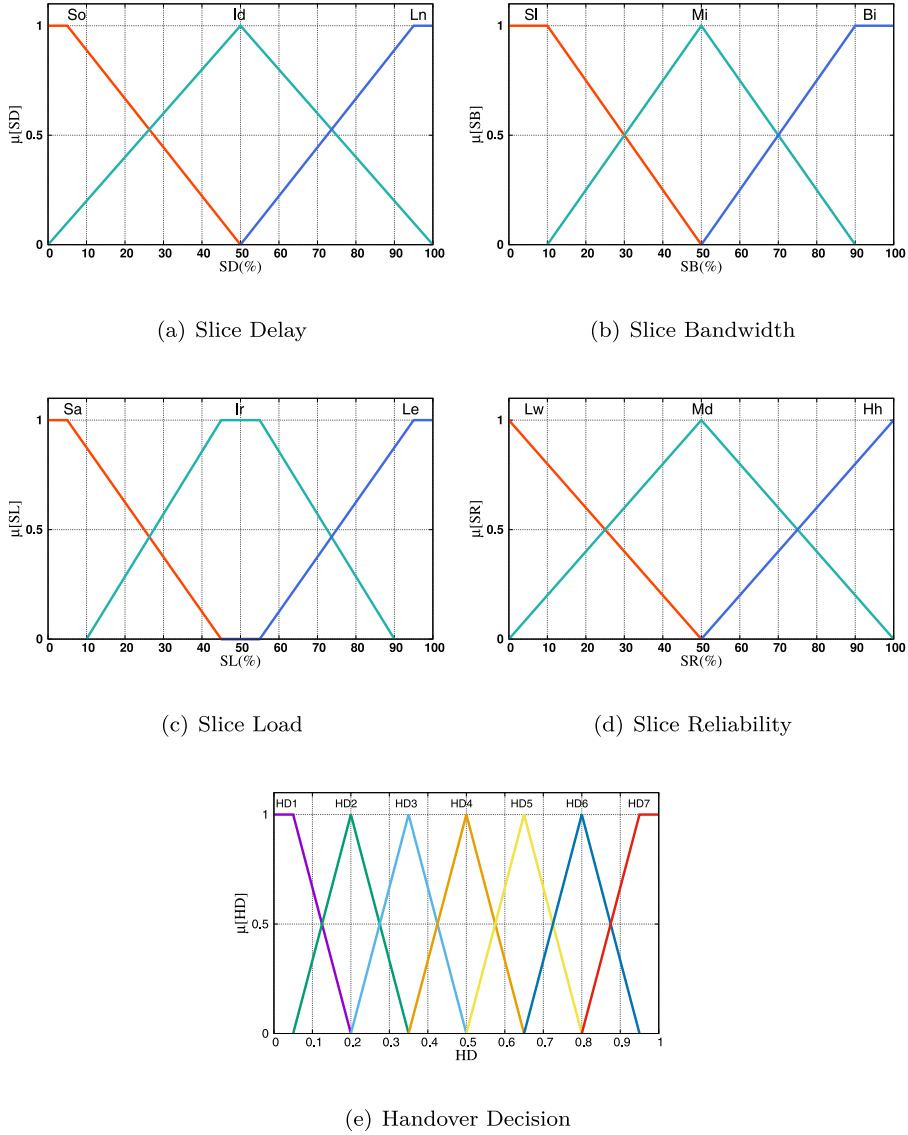


Fig. 8. Membership functions for FHDM1 and FHDM2.

Table 1

Parameter and their term sets for FHDM1 and FHDM2.

Parameters	Term sets
Slice Delay (SD)	Short (So), Intermediate (Id), Long(Ln)
Slice Bandwidth (SB)	Small (Sl), Medium (Mi), Big (Bi)
Slice Load (SL)	Small (Sa), Intermediate (Ir), Large (Le)
Slice Reliability (SR)	Low (Lw), Medium (Md), High (Hh)
Handover Decision (HD)	HD1, HD2, HD3, HD4, HD5, HD6, HD7

sets in Table 1. The Fuzzy Rule Base (FRB) for FHDM1 is shown in Table 2 and has 27 rules. While, the FRB for FHDM2 is shown in Table 3 and has 81 rules.

Table 2
FRB for FHDM1.

Rule	SD	SB	SL	HD	Rule	SD	SB	SL	HD	Rule	SD	SB	SL	HD
1	So	Sl	Sa	HD3	10	Id	Sl	Sa	HD4	19	Ln	Sl	Sa	HD5
2	So	Sl	Ir	HD4	11	Id	Sl	Ir	HD5	20	Ln	Sl	Ir	HD6
3	So	Sl	Le	HD5	12	Id	Sl	Le	HD6	21	Ln	Sl	Le	HD7
4	So	Mi	Sa	HD2	13	Id	Mi	Sa	HD3	22	Ln	Mi	Sa	HD4
5	So	Mi	Ir	HD3	14	Id	Mi	Ir	HD4	23	Ln	Mi	Ir	HD5
6	So	Mi	Le	HD4	15	Id	Mi	Le	HD5	24	Ln	Mi	Le	HD6
7	So	Bi	Sa	HD1	16	Id	Bi	Sa	HD2	25	Ln	Bi	Sa	HD3
8	So	Bi	Ir	HD2	17	Id	Bi	Ir	HD3	26	Ln	Bi	Ir	HD4
9	So	Bi	Le	HD3	18	Id	Bi	Le	HD4	27	Ln	Bi	Le	HD5

Table 3
FRB for FHDM2.

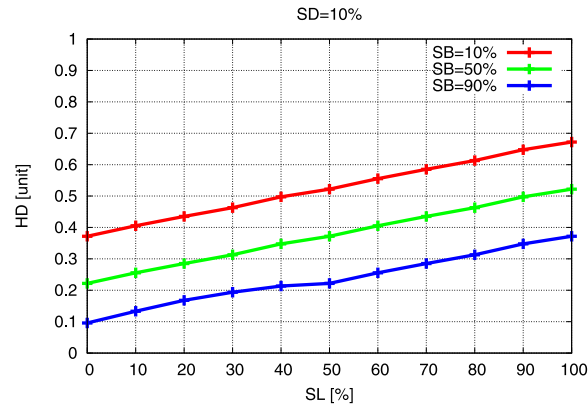
Rule	SD	SB	SL	SR	HD	Rule	SD	SB	SL	SR	HD
1	So	Sl	Sa	Lw	HD4	41	Id	Mi	Ir	Md	HD4
2	So	Sl	Sa	Md	HD3	42	Id	Mi	Ir	Hh	HD3
3	So	Sl	Sa	Hh	HD2	43	Id	Mi	Le	Lw	HD6
4	So	Sl	Ir	Lw	HD5	44	Id	Mi	Le	Md	HD5
5	So	Sl	Ir	Md	HD4	45	Id	Mi	Le	Hh	HD4
6	So	Sl	Ir	Hh	HD3	46	Id	Bi	Sa	Lw	HD3
7	So	Sl	Le	Lw	HD6	47	Id	Bi	Sa	Md	HD2
8	So	Sl	Le	Md	HD5	48	Id	Bi	Sa	Hh	HD1
9	So	Sl	Le	Hh	HD4	49	Id	Bi	Ir	Lw	HD4
10	So	Mi	Sa	Lw	HD3	50	Id	Bi	Ir	Md	HD3
11	So	Mi	Sa	Md	HD2	51	Id	Bi	Ir	Hh	HD2
12	So	Mi	Sa	Hh	HD1	52	Id	Bi	Le	Lw	HD5
13	So	Mi	Ir	Lw	HD4	53	Id	Bi	Le	Md	HD4
14	So	Mi	Ir	Md	HD3	54	Id	Bi	Le	Hh	HD3
15	So	Mi	Ir	Hh	HD2	55	Ln	Sl	Sa	Lw	HD6
16	So	Mi	Le	Lw	HD5	56	Ln	Sl	Sa	Md	HD5
17	So	Mi	Le	Md	HD4	57	Ln	Sl	Sa	Hh	HD4
18	So	Mi	Le	Hh	HD3	58	Ln	Sl	Ir	Lw	HD7
19	So	Bi	Sa	Lw	HD2	59	Ln	Sl	Ir	Md	HD6
20	So	Bi	Sa	Md	HD2	60	Ln	Sl	Ir	Hh	HD5
21	So	Bi	Sa	Hh	HD1	61	Ln	Sl	Le	Lw	HD7
22	So	Bi	Ir	Lw	HD3	62	Ln	Sl	Le	Md	HD7
23	So	Bi	Ir	Md	HD2	63	Ln	Sl	Le	Hh	HD6
24	So	Bi	Ir	Hh	HD1	64	Ln	Mi	Sa	Lw	HD5
25	So	Bi	Le	Lw	HD4	65	Ln	Mi	Sa	Md	HD4
26	So	Bi	Le	Md	HD3	66	Ln	Mi	Sa	Hh	HD3
27	So	Bi	Le	Hh	HD2	67	Ln	Mi	Ir	Lw	HD6
28	Id	Sl	Sa	Lw	HD5	68	Ln	Mi	Ir	Md	HD5
29	Id	Sl	Sa	Md	HD4	69	Ln	Mi	Ir	Hh	HD4
30	Id	Sl	Sa	Hh	HD3	70	Ln	Mi	Le	Lw	HD7
31	Id	Sl	Ir	Lw	HD6	71	Ln	Mi	Le	Md	HD6
32	Id	Sl	Ir	Md	HD5	72	Ln	Mi	Le	Hh	HD5
33	Id	Sl	Ir	Hh	HD4	73	Ln	Bi	Sa	Lw	HD4
34	Id	Sl	Le	Lw	HD7	74	Ln	Bi	Sa	Md	HD3
35	Id	Sl	Le	Md	HD6	75	Ln	Bi	Sa	Hh	HD2
36	Id	Sl	Le	Hh	HD5	76	Ln	Bi	Ir	Lw	HD5
37	Id	Mi	Sa	Lw	HD4	77	Ln	Bi	Ir	Md	HD4
38	Id	Mi	Sa	Md	HD3	78	Ln	Bi	Ir	Hh	HD3
39	Id	Mi	Sa	Hh	HD2	79	Ln	Bi	Le	Lw	HD6
40	Id	Mi	Ir	Lw	HD5	80	Ln	Bi	Le	Md	HD5
						81	Ln	Bi	Le	Hh	HD4

8. Simulation results

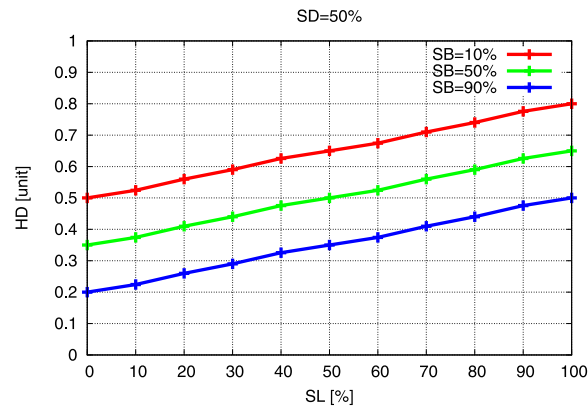
In this section, we present the simulation results. We use a PC with these specifications: Processor i5-3570 (3.2 GHz), RAM (8 GB) and SSD (650 GB). The PC runs Linux Ubuntu OS. We performed simulations using FuzzyC, which is a simulation program written in C and uses Fuzzy library [45].

8.1. Simulation results of FHDM1

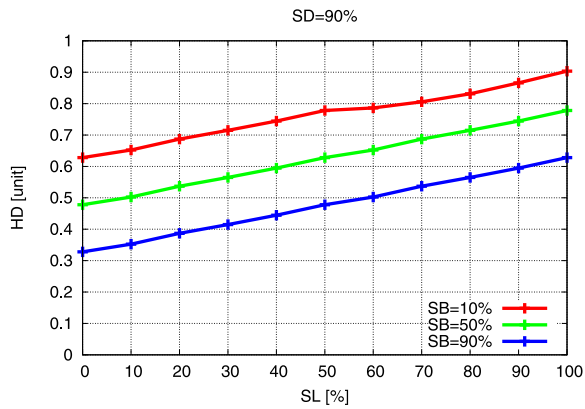
The simulation results of FHDM1 are shown in Fig. 9. They show the relation of HD with SD, SB and SL. The SD is considered as a constant parameter. We increase the SB and SL values from 10% to 90% and 0 to 100%, respectively. In Fig. 9(a), we see that



(a) SD=10%



(b) SD=50%



(c) SD=90%

Fig. 9. Simulation results of FHDM1.

HD is increased when SB is decreasing. The HD value is more than 0.5 when SB is 10% and SL value is more than 40%. This means that the UE is connecting to a slice with a high traffic load and less bandwidth, which will have a high handover possibility for improving service performance. In Fig. 9(b), we increase the SD to 50%. Comparing with Fig. 9(a), we see that HD is increasing by 13% when SL is 50% and SB is 50%. Therefore, the UE needs to find other slices with lower delay to provide better QoS. When SB

value is 10%, the HD is more than 0.5. So, UE has a high possibility to make handover to another slice when the present slice has low bandwidth and intermediate delay. However, when SD is increased to 90% in Fig. 9(c), the HD increases much more compared with Fig. 9(a) and 9(b). In case when SB value is 90%, the HD is greater than 0.5, when SD value is 90% and SL value is greater than 60%.

8.2. Simulation results of FHDM2

The simulation results for FHDM2 are shown in Figs. 10–12. They show the relation of HD with SR, which is considered as a new parameter. We consider SD and SB as constant parameters and SL has different values. In Fig. 10, the SD value is 10% as constant parameter and SB has different values: 10%, 50%, 90% (see Fig. 10(a), 10(b) and 10(c)). When SB value is changed from 10% to 50% and 50% to 90%, the HD values are increased by 15% when SL is 50% and SR is 50%. For low-delay, medium-load and medium-reliability scenarios, the larger is the size of the bandwidth, the low is the handover probability. If SL value is increased to 90% and SR value is less than 50%, the HD value becomes more than 0.5. This indicates that even if the present slice has a larger bandwidth size, the user should make handover to have better QoS (see Fig. 10(a) and 10(b)). In Fig. 10(c), we see that the present slice has big bandwidth and low delay so there is not need for handover.

We increased the SD value from 10% to 50% in Fig. 11. When we compare Fig. 11(a) with Fig. 10(a), the HD is increased by 13% when SB is 10%, SL is 50% and SR is 50%. This means the present slice with intermediate delay has a higher possibility to make handover compared with the slice with lower delay. In Fig. 11(a), the present slice with low bandwidth size (SB is 10%) has all HD values greater than 0.5, when SL values are 50% and 90%. In Fig. 11(b), the HD value is increased by 12% for SR 50%, when SL values are increased from 10% to 50% and 50% to 90%. Comparing Fig. 11(c) with Fig. 11(a) and 11(b), the HD value is decreased by 15% when SR is 80% and SL is 90%, respectively.

We increased the SD value to 90%, as shown in Fig. 12. Comparing Figs. 10 and 11, we can see that the HD values are increased. In Fig. 12(a), all HD values are more than 0.5. This means that when the present slice has a high delay and low bandwidth, the user has a high possibility to make handover to another slice. Comparing Fig. 12(a) with Fig. 10(c), the HD value is increased by 50% when SR is 50% and SL is 90%. This indicates that when a user is connecting with the present slice that has bad QoS, it should handover to another slice with better QoS. In Fig. 12(b) and 12(c), we increased the SB value to 50% and 90%. We see that HD values are decreased compared with Fig. 12(a).

8.3. Discussion

In this subsection, we discuss the simulation results and make a qualitative evaluation considering related works summarized in Tables 4 and 5.

In [11], the authors implement a FL-based algorithm combining with the Multipath Transmission Protocol (MPTCP) in SDN controller. They consider three parameters for handover decision: RSS, data transmission rate, and battery charge. They claim that the proposed algorithm can achieve a seamless handover, avoids the ping-pong effect and reduces unnecessary handovers. However, as the result of handover decision, a UE may handover to another BS, but the destination BS cannot guarantee the same slice QoS to UE.

In [12], in order to increase the QoS, the authors implement FL in SDN controller, which collects QoS measurements and dynamically routes service flows to determine suitable paths. The authors consider two parameters: packet loss rate and delay. However, there are needed more parameters for improving QoS.

In [13], in order to reduce handover delay time and computation, the authors implement a Linear Programming (LP) technique. The proposed technique can satisfy UE with a delay less than 1 ms, but for performing more efficient handover, it needs to consider more parameters.

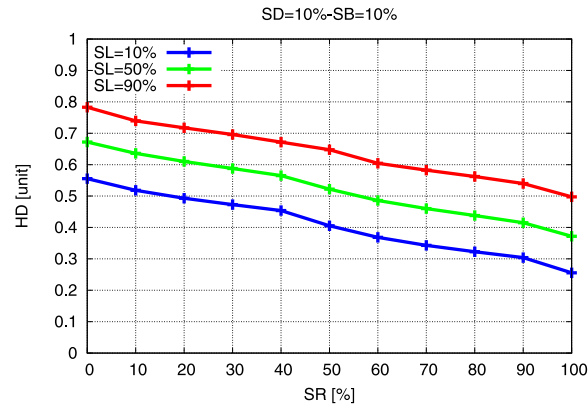
In [14], the simulation results provided by authors show that slice resources utilization is increased, while having low computing time, less number of handovers and less number of dropped slice connections by implementing three heuristics algorithms. However, the proposed heuristics do not provide the optimal solution. They can achieve close-to-optimal solutions.

In [15], the authors use reinforcement learning techniques to minimize handover cost while maintaining user QoS. The proposed approach can improve network signaling exchanges/overhead, number of handovers and outage probability. But, there are needed a lot of data and a high computation time by using reinforcement learning.

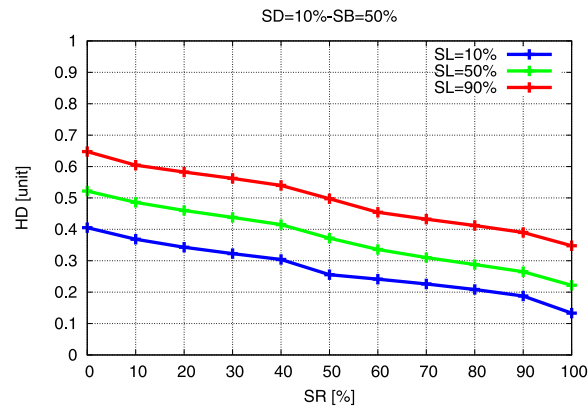
In [16], by assuring that each m-health candidate data flow will be handled, the proposed approach uses a Multiple Attribute Decision Making (MADM) technique to maximize resource consumption by guaranteeing that each m-health candidate data flow will be used effectively. The MADM method is easy to implement, but it cannot deal with human reasoning and high complexity approaches such as Analytic Hierarchy Process (AHP) and rank reversal phenomena. For large networks and other scenarios, there are needed machine learning techniques to determine slice service level requirements more precisely.

For the selection of slices, in [17], the authors present SSF-V2X considering the sigmoid function to evaluate QoS based on the slice service resources such as slice delay, slice bandwidth and slice load. The proposed approach shows better distribution of users among available slices enabling effective resource usage and load balancing provisioning. But, when selecting a slice and evaluating QoS, the reliability and security parameters should be considered.

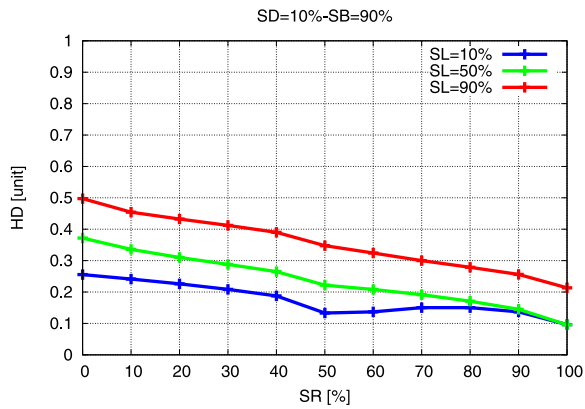
In [18], the authors shows a significant improvement over the traditional RSS based handover scheme (lower handover cost with higher average throughput) by considering handover cost and QoS (Slice Delay and Slice Bandwidth). But, they should consider more parameters for evaluating QoS such as security, reliability and traffic load.



(a) SB=10%



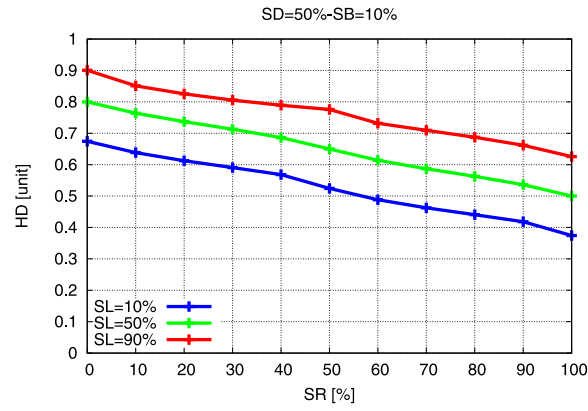
(b) SB=50%



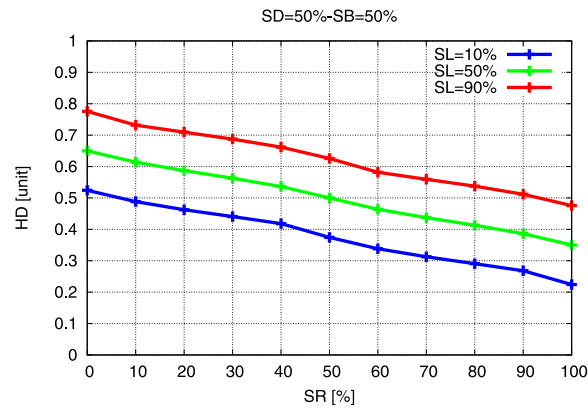
(c) SB=90%

Fig. 10. Simulation results for FHDM2 (SD = 10%).

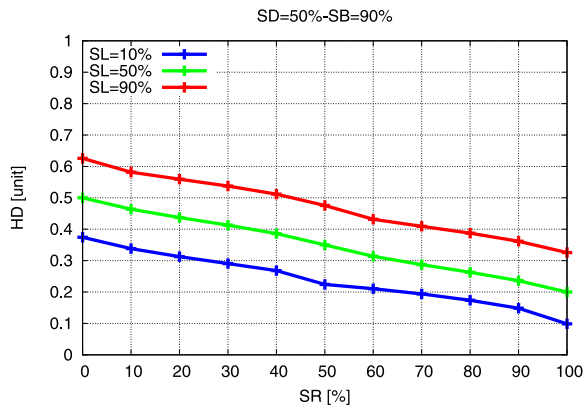
In [19], the authors propose a model that can differentiate between regular congestion and an attack by considering Average Waiting Time (AWT) and Average Switching Rate (ASR). However, they are not considering slice switching patterns and different kinds of attack. Thus, the proposed model cannot predict the attack precisely.



(a) SB=10%



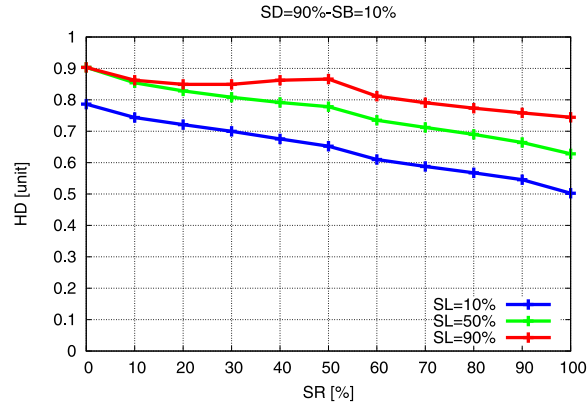
(b) SB=50%



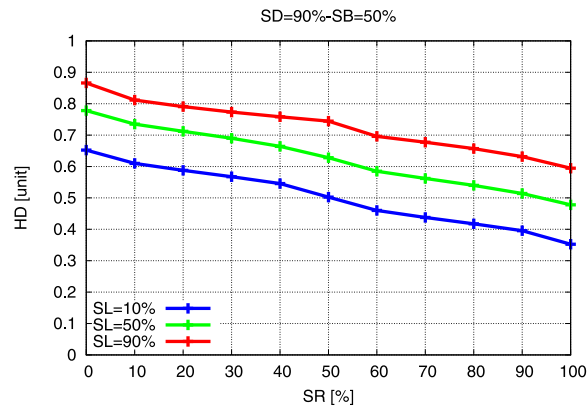
(c) SB=90%

Fig. 11. Simulation results for FHDM2 (SD = 50%).

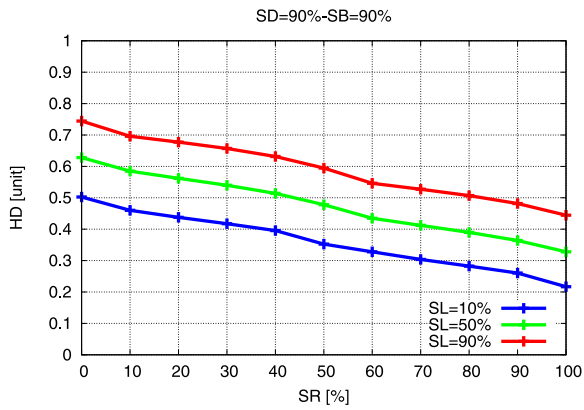
Considering merits and demerits of the abovementioned related works, we have proposed an intelligent system based on FL. In [11,12], the authors propose approaches based on FL. However, in these papers, the authors consider only conventional input parameters such as RSS, data transmission rate, battery charge or packet loss rate and delay. Different from these related works, our research is concentrated on slice parameters for handover decision. We have implemented two models (FHDM1 and FHDM2),



(a) SB=10%



(b) SB=50%



(c) SB=90%

Fig. 12. Simulation results for FHDM2 (SD = 90%).

where FHDM1 considers three input parameters and FHDM2 four input parameters. FHDS2 can provide better handover decision, because it considers 4 input parameters.

Table 4
Summary of related works considering optimized parameters and used approaches.

References	Optimized parameters	Approaches	Year
[11]	RSS, Data Transmission Rate and Battery Power	FL and MPTCP	2018
[12]	Packet Loss Rate, Delay	FL	2018
[13]	Handover Delay Time	LP Technique	2017
[14]	Slices Service Resources (Slice Load)	Three Heuristics Algorithms	2021
[15]	Handover Costs and QoS (Slice Bandwidth)	Reinforcement Learning Techniques (Q-learning)	2020
[16]	Slices Service Resources and QoS	MADM	2022
[17]	Slices Service Resources	SSF- V2X Considering Asigmoid Function	2019
[18]	Handover Cost and QoS (Slice Delay and Slice Bandwidth)	CMaaS and SMaaS	2022
[19]	Slice Security and Reliability	Attack Detection and Localization Algorithms	2023

Table 5
Summary of related research works considering merits and demerits.

References	Merits	Demerits
[11]	Can avoid the ping-pong effect and reduces unnecessary handovers.	Destination BS cannot guarantee the same slice QoS to UE.
[12]	Provide dynamically routes service flows.	Should be considered more parameters.
[13]	Less handover delay time.	More parameters are needed and LP is not suitable for more complex problems.
[14]	Slice resources utilization is increased, while having low computing time, less number of handovers and dropped slice connections.	The proposed heuristics do not provide the optimal solution. They can achieve close-to-optimal solutions.
[15]	Network signaling exchanges/overhead, number of handovers and outage are improved.	Reinforcement learning needs a lot of data and computation time. Thus, it can lead to an overload of states, which can diminish the results.
[16]	Maximizes slice resource utilization and provides slice services.	MADM method cannot deal with non-mathematic approaches and has high complexity
[17]	Better distribution of users among available slices enabling effective resource usage and load balancing.	Needs to consider more parameters for evaluating QoS and selecting slices.
[18]	Significant improvement compared with traditional RSS-based handover scheme.	Needs to consider more parameters for evaluating QoS such as security and reliability
[19]	Can differentiate regular congestion from an attack.	Not considering slice switching patterns and different kind of attacks.

9. Conclusions and future work

In this paper, we presented a Fuzzy-based system for handover in 5G wireless networks considering different network slicing parameters. We implemented two models: FHDM1 and FHDM2. We evaluated the proposed models by simulations. From the simulations results, we conclude as follows.

- The FHDM2 is more complex than FHDM1, but it has better HD since it has four parameters.
- For both models, the HD value increases as SD and SL values increase while HD value is decreasing when SB and SR values increase.
- For better QoS, a mobile device (user) connected to the present slice, which has a high latency, low bandwidth, heavy load, and a low of reliability should to make handover to another slice. For example, in the results of FHDM2, the HD value is more over than 86% when SD is 90%, SB is 10%, SR is 10% and SL is 90%.

In future work, we will consider other parameters and perform extensive simulations to evaluate the effectiveness of the proposed system. We also will compare the performance of the proposed system with other systems. Furthermore, we plan to implement a testbed and compare the simulation results with experimental results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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